13/12/24 M/S

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# FIFTH SEMESTER B. Tech EXAMINATION, DECEMBER-2024

## SOFTWARE AIDED CONSTRUCTION PLANNING AND MANAGEMENT

**Time: 3 Hours** **Maximum Marks: 100**

**Instructions:** i. Missing data may be suitably assumed.

**ANSWER ALL QUESTIONS**

| | | PART-A | 5 x2=10 | | |
|---|---|---|---|---|---|
| | | | BTL | COs | Marks |
| 1. | a. | Describe the main objectives and principles of project planning. | 2 | 1 | 02 |
| | b. | What is the critical path in a project network? How do you identify it using CPM? | 2 | 2 | 02 |
| | c. | Explain the difference between resource smoothing and resource leveling in project management. | 2 | 3 | 02 |
| | d. | How does ISO certification contribute to quality control in a project? | 2 | 4 | 02 |
| | e. | What factors would you consider in a cost optimization problem using project networks? | 2 | 3,6 | 02 |
| | | **PART-B** | **6 X 5=30** | | |
| 2. | a. | Explain the principles of planning in project management and how they contribute to the successful completion of a project. | 2 | 1 | 05 |
| | | **OR** | | | |
| | b. | Describe the difference between Direct Cost and Indirect Cost in a project and give examples of each. | 2 | 1 | 05 |
| | | | | | |
| 3. | a. | How do you determine the Critical Path in a project network diagram? Explain with an example. | 2 | 2 | 05 |
| | | **OR** | | | |
| | b. | Construct the network for the following situation. | 3 | 3 | 05 |

| Activity | I – Node | J – Node |
|---|---|---|
| A | 1 | 2 |
| B | 1 | 3 |
| C | 1 | 4 |
| D | 2 | 5 |
| E | 3 | 5 |
| F | 5 | 6 |
| G | 4 | 6 |
| H | 6 | 7 |
| I | 4 | 7 |

| | | | | | |
|---|---|---|---|---|---|
| | | | | | |
| 4. | a. | Explain how you would optimize cost using project networks. Provide steps you would take to minimize the total project cost. | 2 | 3 | 05 |
| | | **OR** | | | |

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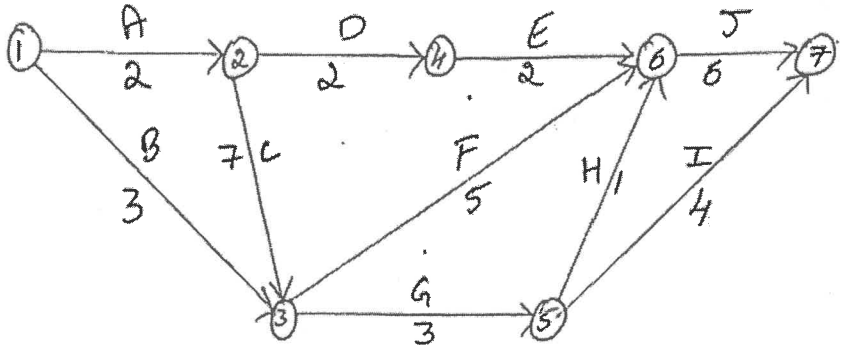
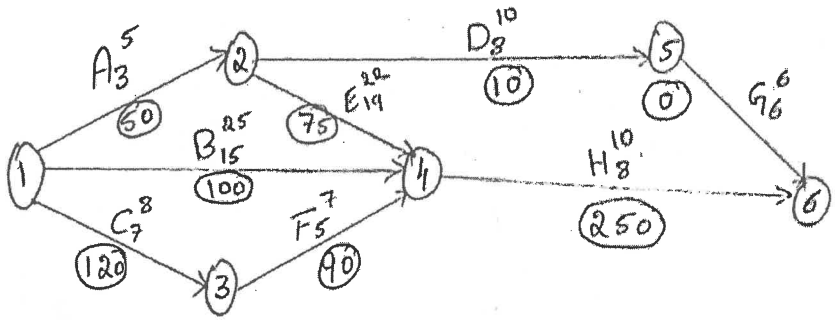
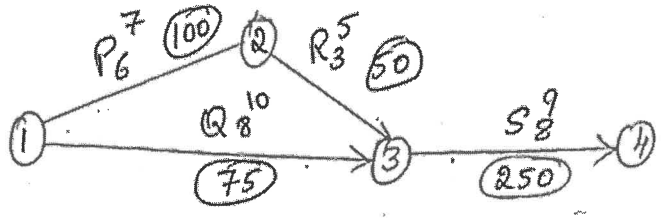
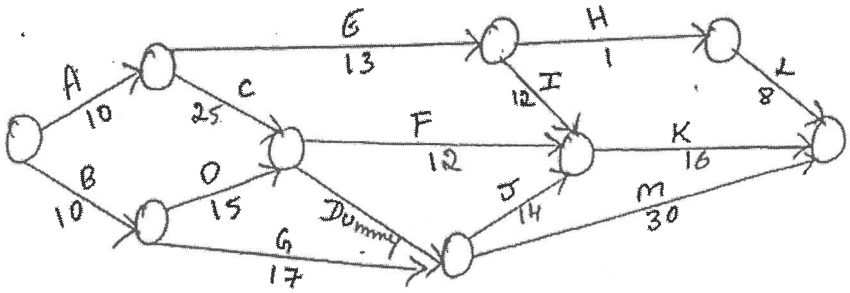
|---|---|---|---|---|---|
| | **b.** | Define resource leveling and describe how it impacts the project schedule. Provide an example. | 2 | 3 | **05** |
| | | | | | |
| **5.** | **a.** | What are the key steps in achieving ISO certification for a project? List some of the requirements needed to ensure compliance with ISO standards. | 2 | 4 | **05** |
| | | **OR** | | | |
| | **b.** | What is the importance of having a checklist during quality inspections in a project? Explain with examples. | 2 | 4 | **05** |
| | | | | | |
| **6.** | **a.** | How would you use project management software like Primavera P6 or MS Projects to track the progress of a real-world project? | 2 | 2,6 | **05** |
| | | **OR** | | | |
| | **b.** | What quality control methods would you recommend for a project, and how do they help ensure project success? | 2 | 4,6 | **05** |
| | | | | | |
| **7.** | **a.** | Find the critical path and project duration of the given CPM Project (Figure. 7a).   **Figure. 7a** | **3** | **3** | **05** |
| | | **OR** | | | |
| | **b.** | How would you prepare a final project report, and what key aspects would you include to communicate project results effectively? | 2 | 5,6 | **05** |
| | | **PART-C** | **6 X 10=60** | | |
| **8.** | **a.** | Draw a project network diagram for the following activities and calculate the Critical Path and project duration using the Critical Path Method (CPM): | 3 | 2 | **10** |

| Activity | Predecessor | Duration (days) |
|---|---|---|
| A | - | 4 |
| B | A | 6 |
| C | A | 5 |
| D | B | 4 |
| E | B,C | 3 |

| | | | | | |
|---|---|---|---|---|---|
| | | **OR** | | | |

| | | | | | |
|---|---|---|---|---|---|
| | **b.** | Explain the steps involved in creating a project schedule using Primavera P6. How would you track project progress using this software? | 2 | 2,6 | **10** |
| **9.** | **a.** | A small project consist s of activities listed below. if the indirect cost is rs. 200rs/day and total direct cost is 25,500 rs, what is the optimum and minimum (all crash) duration of the project (Figure. 9a)?   **Figure. 9a** | **4** | **3** | **10** |
| | | **OR** | | | |
| | **b.** | Explain the significance of Earned Value Management (EVM) in project management. How does EVM help in assessing project performance? | 2 | 2 | **10** |
| | | | | | |
| **10.** | **a.** | A small project consists of activities shown below. if the indirect cost is Rs. 190/day and total direct cost is 1600rs. what is the normal, optimum and minimum duration of the project (Figure 10a)?   **Figure. 10a** | **4** | **3** | **10** |
| | | **OR** | | | |
| | **b.** | Explain the concept of resource leveling in project Management. How does resource leveling affect the project duration and cost? | 3 | 3 | **10** |
| **11.** | **a.** | Describe the process of obtaining ISO certification for a project. What are the benefits of ISO certification in terms of quality management? | 4 | 4 | **10** |
| | | **OR** | | | |
| | **b.** | What key points should be included in a quality inspection checklist during project execution? Explain how a well-prepared checklist ensures project quality. | 2 | 4 | **10** |
| | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| **12.** | **a.** | Using a real-world project example, explain how Primavera P6 can be used to manage the entire project lifecycle from planning to reporting. | 2 | 6 | **10** |
| | | **OR** | | | |
| | **b.** | How would you implement quality control measures in a construction project? Explain the tools and techniques you would use. | 2 | 4,6 | **10** |
| | | | | | |
| **13.** | **a.** | Calculate all activity times and all float times of the following CPM network and hence obtain critical path and project duration (Figure. 13a).   **Figure. 13a** | **3** | **3** | **10** |
| | | **OR** | | | |
| | **b.** | What are the key differences between CPM (Critical Path Method) and PERT (Program Evaluation Review Technique)? Explain when each should be used in a project. | 2 | 2,3 | **10** |